

1. Discuss the KDD (Knowledge Discovery from Database) process with proper diagram.

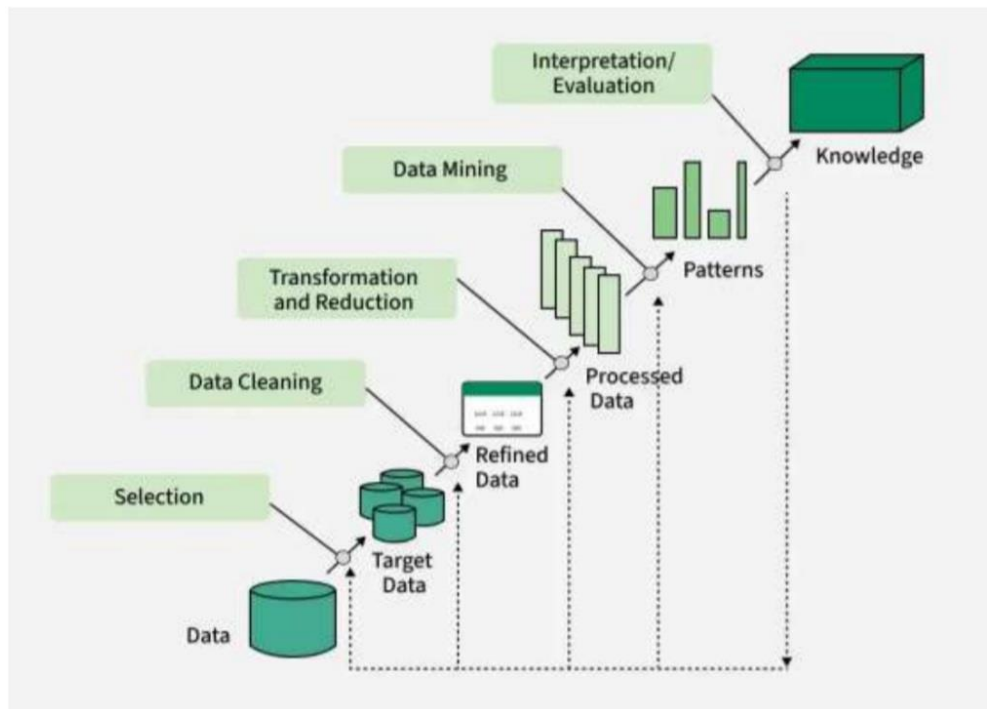
Answer:

The Knowledge Discovery in Databases (KDD) process refers to the broad process of discovering knowledge in data, and emphasizes the "high-level" application of particular data mining methods. It involves several stages:

Steps of KDD:

1. **Data Cleaning:**
This step involves removing noise and irrelevant data. It handles missing values, duplicate records, and inconsistent data entries to improve data quality.
2. **Data Integration:**
Data from multiple sources (e.g., databases, files, or web) are combined into a coherent data store, like a data warehouse. It resolves schema conflicts and redundancy.
3. **Data Selection:**
Only the relevant data to the analysis task are retrieved from the database. This step helps reduce processing load and increases focus.
4. **Data Transformation:**
The selected data are transformed or consolidated into forms appropriate for mining. Techniques like normalization, aggregation, and discretization are used.
5. **Data Mining:**
This is the core step where intelligent methods are applied to extract data patterns. Techniques include classification, clustering, association rule mining, etc.
6. **Pattern Evaluation:**
The patterns discovered are evaluated for their interestingness. Measures like support, confidence, lift, and statistical tests are used to validate significance.
7. **Knowledge Presentation:**
The final step involves visualization and user-friendly representation of mined knowledge. Tools like charts, decision trees, and graphs are used.

Diagram of KDD Process:



2. Outline the benefits of Data Warehouse.

Answer:

A Data Warehouse (DW) is a centralized repository designed to support decision-making by providing clean, integrated, historical data. Key benefits include:

Benefits of Data Warehouse

Better business analytics: Data warehouse plays an important role in every business to store and analysis of all the past data and records of the company. which can further increase the understanding or analysis of data for the company.

Faster Queries: The data warehouse is designed to handle large queries that's why it runs queries faster than the database.

Improved data Quality: In the data warehouse the data you gathered from different sources is being stored and analyzed it does not interfere with or add data by itself so your quality of data is maintained and if you get any issue regarding data quality then the data warehouse team will solve this.

Historical Insight: The warehouse stores all your historical data which contains details about the business so that one can analyze it at any time and extract insights from it.

3. What is Indexing & Querying in OLAP?

Answer:

OLAP (Online Analytical Processing) enables users to analyze multidimensional data interactively from multiple perspectives.

Indexing in OLAP:

- **Definition:**
Indexing in OLAP is a technique used to improve the efficiency of querying and retrieval operations on large multidimensional datasets.
- **Types:**
 - **Bitmap Indexing:** Used for low-cardinality columns, where each bit represents the existence of a value.
 - **B-tree Indexing:** For high-cardinality data.
 - **Join Index:** Speeds up join operations between fact and dimension tables.
- **Benefits:**
 - Accelerates data retrieval.
 - Reduces disk I/O.
 - Optimizes aggregation functions.

Querying in OLAP:

- **Definition:**
Querying is the process of retrieving relevant data using multidimensional expressions (MDX) or SQL-like queries.
- **Operations:**
 - **Slice:** Selects a single dimension from a cube.
 - **Dice:** Selects data based on multiple dimensions.
 - **Drill Down/Up:** Navigates among levels of data ranging from most summarized (up) to most detailed (down).
 - **Pivot (Rotate):** Reorients the multidimensional view of data.
- **Example:**
Query: “Show total sales in 2023 for Electronics in California”
This might use slice (on year), dice (on product & region), and roll-up (on sales).

4. Differentiate OLAP & OLTP.

Answer:

Differentiate between OLAP and OLTP

Sr. No.	Key	OLAP	OLTP
1	Basic	It is used for data analysis	It is used to manage very large number of online short transactions
2	Database Type	It uses data warehouse	It uses traditional DBMS
3	Data Modification	It manages all insert, update and delete transaction	It is mainly used for data reading
4	Response time	Processing is little slow	In Milliseconds
5	Normalization	Tables in OLAP database are not normalized.	Tables in OLTP database are normalized.

5. Explain various schema involved in conceptual modelling of a data warehouse.

Answer:

Conceptual modeling in data warehousing involves designing schemas that define how data is organized. The most common schemas are:

1. **Star Schema:**
 - Central fact table connected to dimension tables.
 - Easy to understand and optimize for OLAP queries.
 - Example: Sales Fact Table connected to Product, Time, Region dimensions.
2. **Snowflake Schema:**
 - A variation of the star schema where dimension tables are normalized.
 - Reduces data redundancy but is complex and slower for query processing.

3. **Fact Constellation (Galaxy Schema):**
 - Multiple fact tables share dimension tables.
 - Supports complex and multi-fact analysis.
4. **Starflake Schema:**
 - A hybrid of star and snowflake schema.
 - Some dimensions are normalized while others are not.

6. Discuss the different components of a data warehouse.

Answer:

1. Operational Source Systems

- Provide raw data from internal systems (e.g., relational databases like Informix oracle) and external sources.
- Serve as the primary input for the data warehouse.

2. Load Manager

- Manages the ETL (Extract, Transform, Load) process for data extraction and transformation.
- Prepares data for entry into the warehouse, ensuring it meets the required format.

3. Warehouse Manager

- Oversees data storage, aggregation and analysis within the data warehouse.
- Handles tasks like data de-normalization, backup, collection and optimization for better performance.

4. Query Manager

- Handles user queries within the data warehouse.
- Supports querying, reporting and data retrieval, with functionality dependent on the available end-user tools.

5. Detailed Data

- Stores granular, raw data for complex analysis and reporting.
- Provides comprehensive insights and supports in-depth research.

6. Summarized Data

- Stores predefined aggregations of detailed data for faster queries and reports.
- Provides high-level insights to assist in decision-making.

7. Archive and Backup Data

- Ensures data integrity and disaster recovery through regular backups and archival storage.
- Archives older, less frequently accessed data for future use or compliance.

8. Metadata

- Metadata contains information about data structure, source and transformation processes.
- Supports the ETL process, warehouse management and querying by providing essential context for data.

9. End-User Access Tools

- Serve as the interface for users to interact with the data warehouse.
- Include analysis, reporting and data mining tools, enabling users to access, query and derive insights from the data.

7. Define Data Warehouse. Explain multi-tier architecture of data warehouse.

Answer:

A **Data Warehouse** is a centralized, subject-oriented, integrated, time-variant, and non-volatile collection of data that supports decision-making processes.

Multi-Tier Architecture:

Bottom Tier(Data sources and data storage) :

The bottom Tier usually consists of Data Sources and Data Storage.

It is a warehouse database server. For Example RDBMS.

In Bottom Tier, using the application program interface(called gateways), data is extracted from

operational and external sources.

Application Program Interface likes ODBC(Open Database Connection), OLE-DB(Open-Linking and

Embedding for Database), JDBC(Java Database Connection) is supported.

ETL stands for Extract, Transform, and Load.

Several popular ETL tools include:

I. IBM Infosphere

II. Informatica

III. Confluent

IV. Microsoft SSIS

V. Snaplogic

VI. Alooma

Middle Tier :

The middle tier is an OLAP server that is typically implemented using either :

A relational OLAP (ROLAP) model (i.e., an extended relational DBMS that maps operations from standard data to standard data); or A multidimensional OLAP (MOLAP) model (ie, a special purpose server that directly implements multidimensional data and operations).

OLAP server models come in three different categories, including:

- 1. ROLAP: A relational database is not converted into a multidimensional database; rather, a relational database is actively broken down into several dimensions as part of relational online analytical processing(ROLAP). This is used when everything that is contained in the repository is a relational database system.**
- 2. MOLAP: A different type of online analytical processing called multidimensional online analytical processing(MOLAP) includes directories and catalogs that are immediately integrated into its multidimensional database system. This is used when all that is contained in the repository is the multidimensional database system.**
- 3. HOLAP: A combination of relational and multidimensional online analytical processing paradigms is hybrid online analytical processing(HOLAP). HOLAP is the ideal option for a seamless functional flow across the database systems when the repository houses both the relational database management system and the multidimensional database management system.**

Top Tier :

The top tier is a front-end client layer, which includes query and reporting tools, analysis tools, and/or data mining tools (eg, trend analysis, prediction, etc.).

Here are a few Top Tier tools that are often used:

I. SAP BW

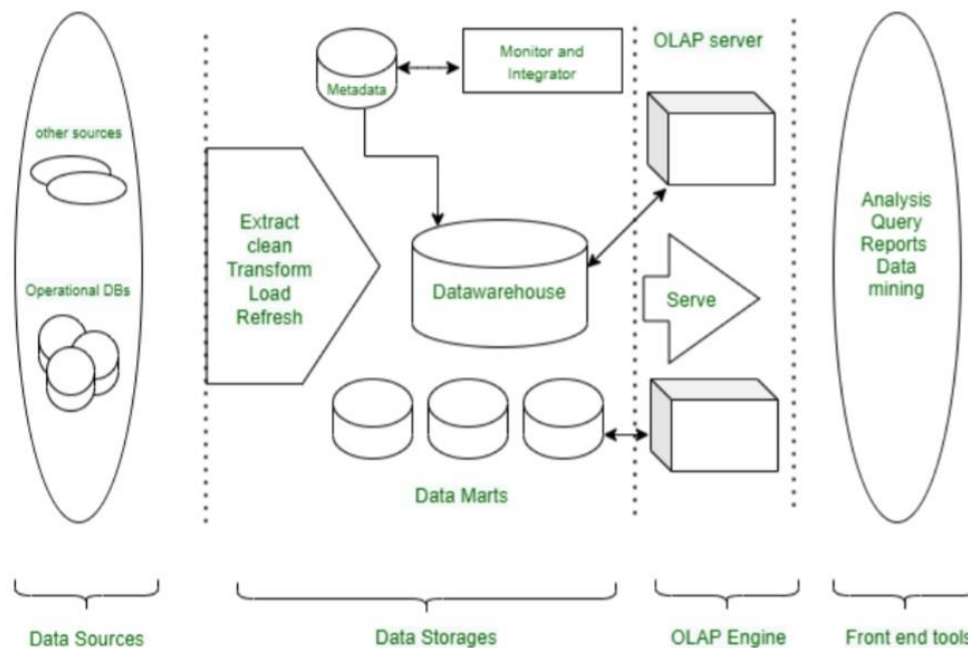
II. SAS Business Intelligence

III. IBM Cognos

IV. Crystal Reports

V. Microsoft BI Platform

Architecture Diagram:



8. Define Data Warehouse and Data Mart.

Answer:

- **Data Warehouse:**

A centralized repository that stores integrated and historical data from different sources to support decision-making. It provides a unified platform for querying and analysis.

Data Mart:

Data Mart:-

A data mart contains a subset of corporate-wide data that is important to a specific group of users.

The scope is limited to specific selected subjects.

For example, a marketing data mart may limit its topics to customers, goods, and sales.

The data contained in the data marts are summarized. Data marts are typically applied to low-cost departmental servers that are Unix/Linux or Windows-based.

The implementation cycle of a data mart is more likely to be measured in weeks rather than months or years. However, in the long run, complex integration is involved in its design and planning were not enterprise-wide.

Differences:

Feature	Data Warehouse	Data Mart
Scope	Enterprise-wide	Department-level

Feature	Data Warehouse	Data Mart
Size	Large	Smaller
Data Source	Multiple	Few or single source
Complexity	High	Relatively low
Implementation Time	Long	Short

9. What is the use of metadata in data warehouse?

Answer:

Metadata is "data about data" and plays a crucial role in the functioning of a data warehouse. It provides information about the structure, operations, and semantics of data.

Uses of Metadata:

1. **Data Lineage:** Shows where data originated from and how it has changed.
 2. **Data Transformation Logic:** Defines ETL rules and mappings.
 3. **Schema Definitions:** Helps tools understand the structure of tables and relationships.
 4. **Query Optimization:** Assists in formulating efficient query plans.
 5. **User Guidance:** Aids users in understanding how to access and interpret data.
 6. **Data Governance:** Ensures compliance with policies and standards.
-

10. What are the major tasks done in data pre-processing?

Answer:

Data pre-processing is a crucial step in data mining, aiming to improve the quality of the data and make it suitable for analysis.

Major Tasks:

1. **Data Cleaning:**
Handles missing values, removes duplicates, and corrects inconsistencies.
2. **Data Integration:**
Combines data from multiple sources into a coherent store.
3. **Data Transformation:**
Applies normalization, discretization, and aggregation.
4. **Data Reduction:**
Reduces data volume using dimensionality reduction, numerosity reduction, and data compression.
5. **Data Discretization and Binarization:**
Converts continuous data into categorical and binary forms.
6. **Noise Removal:**
Filters out outliers and random errors using statistical techniques.

Pre-processing ensures the mining results are accurate, meaningful, and efficient.

11. Explain the 3-tier data warehouse architecture and its various components.

Same as 7

12. Explain the ETL process in data warehousing and outline its need.

Answer:

ETL stands for **Extract, Transform, Load**—a critical process for building data warehouses.

1. Extraction:

The first step of the ETL process is extraction. In this step, data from various source systems is extracted which can be in various formats like relational databases, No SQL, XML, and flat files into the staging area. It is important to extract the data from various source systems and store it into the staging area first and not directly into the data warehouse because the extracted data is in various formats and can be corrupted also. Hence loading it directly into the data warehouse may damage it and rollback will be much more difficult. Therefore, this is one of the most important steps of ETL process.

2. Transformation:

The second step of the ETL process is transformation. In this step, a set of rules or functions are applied on the extracted data to convert it into a single standard format. It may involve following processes/tasks:

- ☐ **Filtering** – loading only certain attributes into the data warehouse.
- ☐ **Cleaning** – filling up the NULL values with some default values, mapping U.S.A, United States, and America into USA, etc.
- ☐ **Joining** – joining multiple attributes into one.
- ☐ **Splitting** – splitting a single attribute into multiple attributes.
- ☐ **Sorting** – sorting tuples on the basis of some attribute (generally key-attribute).

3. Loading:

The third and final step of the ETL process is loading. In this step, the transformed data is finally loaded into the data warehouse. Sometimes the data is updated by loading into the data warehouse very frequently and sometimes it is done after longer but regular intervals. The rate and period of loading solely depends on the requirements and varies from system to system.

Need for ETL:

- Maintains data quality and integrity.
- Integrates data from various systems.
- Prepares data for efficient querying and analysis in the warehouse.

13. What is OLAP? Outline the OLAP operations with an example.

Answer:

OLAP (Online Analytical Processing) is a category of software tools that allows users to analyze multidimensional data interactively.

Key OLAP Operations:

9 / 21 | 81%

OLAP

Roll-Up

- Roll-up performs aggregation on a data cube in any of the following ways –
- By climbing up a concept hierarchy for a dimension
- By dimension reduction
- The following diagram illustrates how roll-up works.

Locations (countries): USA, Canada

Time (Quarter): Q1, Q2, Q3, Q4

Mobile Modem Phone Security Item(types)

Locations (cities): Chicago, New York, Toronto, Vancouver

Time (Quarter): Q1, Q2, Q3, Q4

Mobile Modem Phone Security Item(types)

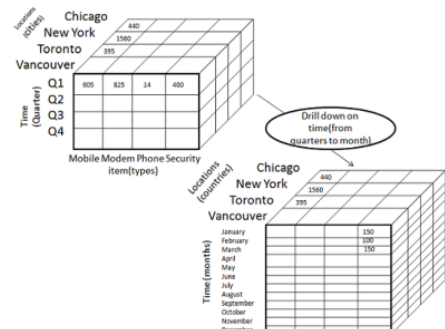
- Roll-up is performed by climbing up a concept hierarchy for the dimension location.
- Initially the concept hierarchy was "street < city < state < country".
- On rolling up, the data is aggregated by ascending the location hierarchy from the level of city to the level of country.
- The data is grouped into cities rather than countries.
- When roll-up is performed, one or more dimensions from the data cube are removed.

Drill-down

- Drill-down is the reverse operation of roll-up. It is performed by either of the following ways –
- The drill-down operation (**also called roll-down**) is the reverse operation of **roll-up**. Drill-down is like **zooming-in** on the data cube. It navigates from less detailed record to more detailed data. By stepping down a concept hierarchy for a dimension
- By introducing a new dimension.

The following diagram illustrates how drill-down works –

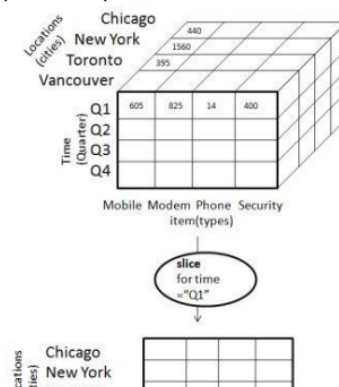
- Drill-down is performed by stepping down a concept hierarchy for the dimension time.
- Initially the concept hierarchy was "day < month < quarter < year."
- On drilling down, the time dimension is descended from the level of quarter to the level of month.
- When drill-down is performed, one or more dimensions from the data cube are added.
- It navigates the data from less detailed data to highly



Slice

- A **slice** is a subset of the cubes corresponding to a single value for one or more members of the dimension. For example, a slice operation is executed when the customer wants a selection on one dimension of a three-dimensional cube resulting in a two-dimensional site. So, the Slice operations perform a selection on one dimension of the given cube, thus resulting in a subcube.
- The slice operation selects one particular dimension from a given cube and provides a new sub-cube. Consider the following diagram that shows how slice works.

- Here Slice is performed for the dimension "time" using the criterion time = "Q1".
- It will form a new sub-cube by selecting one or more dimensions.

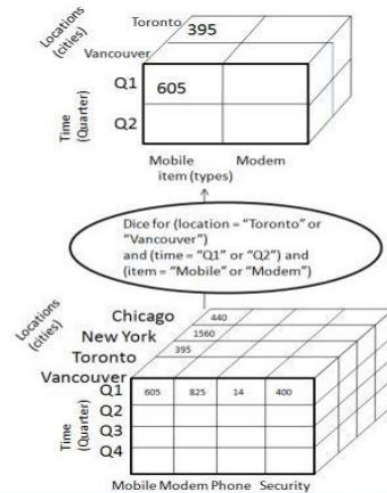


Dice

- The dice operation describes a sub-cube by operating a selection on two or more dimension.
- Dice selects two or more dimensions from a given cube and provides a new sub-cube. Consider the following diagram that shows the dice operation.

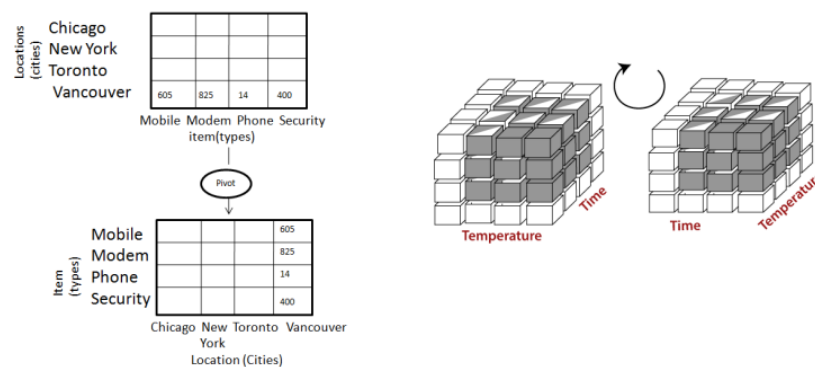
The dice operation on the cube based on the following selection criteria involves three dimensions.

- (location = "Toronto" or "Vancouver")
- (time = "Q1" or "Q2")
- (item = "Mobile" or "Modem")



Pivot

- The pivot operation is also called a rotation. Pivot is a visualization operations which rotates the data axes in view to provide an alternative presentation of the data. It may contain swapping the rows and columns or moving one of the row-dimensions into the column dimensions.
- Consider the following diagram that shows the pivot operation.



14. Outline the schemes for integration of a data mining system with a data warehouse.

Answer:

Integration of data mining systems with data warehouses ensures efficient knowledge discovery. There are **four major integration schemes**:

1. No Coupling:

- Data mining operates independently from data warehouse.
- Data is manually collected, cleaned, and fed into mining tools.

- Not efficient and lacks scalability.

2. Loose Coupling:

- Data mining system accesses warehouse data via OLAP or other middleware.
- Metadata is not shared.
- Limited optimization and performance.

3. Semi-Tight Coupling:

- Partial integration with shared metadata and intermediate results stored in the warehouse.
- Improves efficiency and allows reuse of mining results.

4. Tight Coupling:

- Full integration where data mining functionalities are embedded in the data warehouse system.
- Shares storage, metadata, and processing.
- Most efficient and scalable.

15. Define data mining and explain in detail about the data warehouse architecture with a neat diagram.

Same as 11

16. Propose a supervised learning strategy for predicting spam mail.

Answer:

A supervised learning strategy uses labeled data to train a model to classify emails as “spam” or “not spam”.

Steps:

- 1. Data Collection:**
 - Labeled dataset of emails (with "spam"/"ham" tags).
- 2. Preprocessing:**
 - Tokenization, stopword removal, stemming.
 - Convert text into numeric form using TF-IDF or Bag of Words.
- 3. Feature Selection:**
 - Important features include: frequency of words like “free”, “offer”, “click”, etc.
- 4. Model Selection:**
 - Use classification algorithms like Naïve Bayes, Logistic Regression, or SVM.
- 5. Training and Testing:**
 - Split data (e.g., 80% train, 20% test).

- Train the classifier and evaluate using accuracy, precision, recall.
 - 6. **Deployment:**
 - Use the model in real-time spam filters in email systems.
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20. What do you mean by multilevel association rules? Discuss different variations to this approach.

Answer:

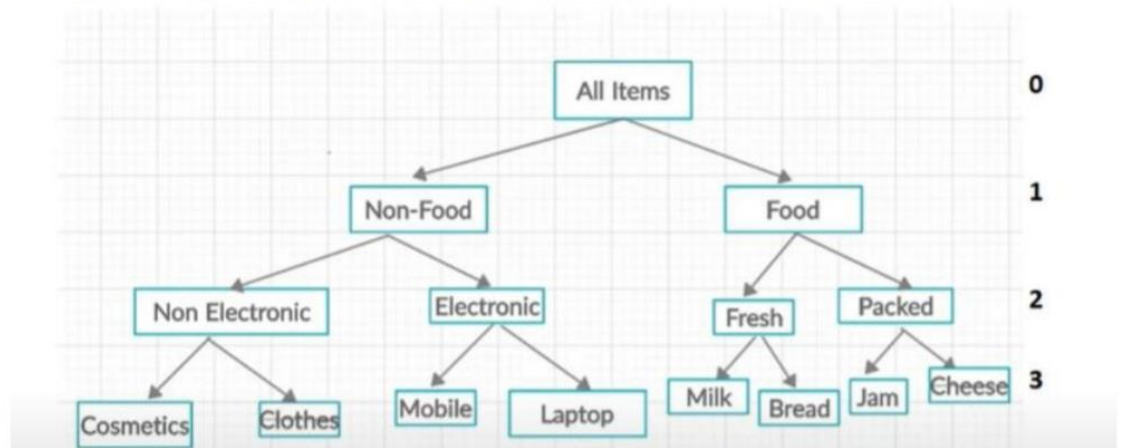
Multilevel Association Rules involve mining rules at multiple levels of abstraction. For example:

- Rule at level 1: "Beverage → Snacks"
- Rule at level 2: "Soft Drink → Chips"

Variations:

1. **Uniform Support:**
 - Same minimum support across all levels.
 - May miss low-frequency but important rules at lower levels.
2. **Reduced Support:**
 - Lower minimum support for deeper levels.
 - Encourages discovery of more specific rules.
3. **Level-by-Level Mining:**
 - Start at top level, then drill-down progressively using mined rules as a base.
4. **Using Concept Hierarchies:**
 - Hierarchical structures (e.g., Electronics → Mobiles → Smartphones) guide rule generalization/specialization.
5. **Cross-Level Rules:**
 - Rules involving items at different levels (e.g., "Soft Drink → Food")

Concept Hierarchy: It is a sequence of mappings from a set of low-level concepts to higher level, more general concepts. Below figure has five levels from levels 0 to 4,



Need of Multiple-Level Association Rules?

- Sometimes at low data level, data does not show any significant pattern. But there are useful information hiding behind.
- Aim is to find the hidden information in or between levels of abstraction

21. Discuss about the FP-Growth Algorithm.

Answer:

FP-Growth (Frequent Pattern Growth) is a highly efficient algorithm for mining frequent itemsets without candidate generation.

Steps:

1. **Construct FP-Tree:**
 - Scan DB once to count item frequencies.
 - Discard infrequent items (below minsup).
 - Sort frequent items and build tree where paths represent transactions.
2. **Mine the FP-Tree:**
 - Starting from each frequent item, construct conditional pattern base.
 - Build conditional FP-tree and mine it recursively.

Advantages:

- Faster than Apriori (no candidate generation).
- Requires only two scans of the database.
- Efficient in large datasets.

Limitations:

- May not fit in memory for large and dense datasets.
-

22. What are multidimensional association rules? Mention few approaches to mining multilevel association rules.**Answer:**

Multidimensional Association Rules are rules involving multiple dimensions (attributes) in a dataset.

Example:

- “Age=20-30 $\wedge$ Income=High $\rightarrow$ Buy=Car”

Approaches to Mining Multilevel Association Rules:

1. **Apriori-Based Approach:**
 - Extend Apriori to handle itemsets with multiple dimensions.
 2. **Classification-Based:**
 - Use concept hierarchies for dimensions and mine rules from different abstraction levels.
 3. **Using Cube Structure:**
 - Use OLAP cubes to identify interesting multidimensional relationships.
 4. **Quantitative Rules:**
 - Map numeric attributes to categorical intervals before mining.
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23. Explain the dynamic quantitative association rule mining for multidimensional association rules.**Answer:**

Dynamic Quantitative Association Rule Mining discovers rules where quantitative attributes (like age, income) are treated dynamically using data distribution.

Key Aspects:

1. **Dynamic Interval Creation:**

- Instead of fixed intervals, use methods like clustering or decision trees to generate intervals based on data.
- 2. **Support and Confidence Computation:**
 - Similar to standard association rule mining but extended for multidimensional numeric data.
- 3. **Rule Format:**
 - E.g., “Age [25–35] $\wedge$ Salary [50k–70k] $\rightarrow$ Buys=Laptop”
- 4. **Handling Quantitative and Categorical Data Together:**
 - Transforms numeric attributes dynamically and mines with categorical ones.

Benefits:

- More meaningful intervals.
 - Rules adapt to the nature of data.
 - Reduces redundancy and increases precision.
-

33. Write the algorithm of KNN classification concept. What are the advantages of KNN classification approach?

Answer:

K-Nearest Neighbor(KNN) Algorithm for Machine Learning

- K-Nearest Neighbour is one of the simplest Machine Learning algorithms based on Supervised Learning technique.
 - K-NN algorithm assumes the similarity between the new case/data and available cases and put the new case into the category that is most similar to the available categories.
 - K-NN algorithm stores all the available data and classifies a new data point based on the similarity. This means when new data appears then it can be easily classified into a well suite category by using K- NN algorithm.
 - K-NN algorithm can be used for Regression as well as for Classification but mostly it is used for the Classification problems.
 - K-NN is a **non-parametric algorithm**, which means it does not make any assumption on underlying data.
 - It is also called a **lazy learner algorithm** because it does not learn from the training set immediately instead it stores the dataset and at the time of classification, it performs an action on the dataset.
 - KNN algorithm at the training phase just stores the dataset and when it gets new data, then it classifies that data into a category that is much similar to the new data.
 - **Example:** Suppose, we have an image of a creature that looks similar to cat and dog, but we want to know either it is a cat or dog. So for this identification, we can use the KNN algorithm, as it works on a similarity measure. Our KNN model will find the similar features of the new data set to the cats and dogs images and based on the most similar features it will put it in either cat or dog category.
-

How does K-NN work?

The K-NN working can be explained on the basis of the below algorithm:

- **Step-1:** Select the number K of the neighbors
- **Step-2:** Calculate the Euclidean distance of **K number of neighbors**
- **Step-3:** Take the K nearest neighbors as per the calculated Euclidean distance.
- **Step-4:** Among these k neighbors, count the number of the data points in each category.
- **Step-5:** Assign the new data points to that category for which the number of the neighbor is maximum.
- **Step-6:** Our model is ready.

Suppose we have a new data point and we need to put it in the required category. Consider the below image:

34. Explain overfitting and underfitting problem along with some remedial measures.

Answer:

Overfitting:

- Occurs when a model learns both the **patterns and noise** in the training data.
- It performs well on training data but poorly on new, unseen data.
- Symptoms include high training accuracy but low testing accuracy.

Causes:

- Model is too complex.
- Too many parameters or features.
- Insufficient training data.

Remedies:

- Use **cross-validation**.
 - Apply **regularization techniques** (L1/L2).
 - Use **simpler models**.
 - Collect more training data.
 - **Prune** decision trees.
-

Underfitting:

- Occurs when a model is too simple to capture underlying trends.
- Both training and testing performance is poor.

Causes:

- Model lacks complexity.
- Insufficient training time.
- Features are not expressive enough.

Remedies:

- Use **more complex algorithms**.
 - Feature engineering (add informative features).
 - Increase training time.
-

35. What do you mean by Non-Linear Regression? How curve fitting is related with Non-Linear Regression?

Answer:

Non-Linear Regression:

- A regression technique where the relationship between dependent and independent variables **is not linear**.
- Used when data forms a curve rather than a straight line.

General Form:

$$y = f(x) + \epsilon$$

Where **f(x)** is a nonlinear function.

Curve Fitting:

- The process of constructing a curve that best fits a series of data points.
- Non-linear regression is a form of curve fitting where a non-linear model is used to match the data pattern.

Applications:

- Population growth modeling.
- Pharmacokinetics.
- Engineering stress-strain curves.

37. What are the prediction techniques supported by a data mining system?

Answer:

Data mining supports several prediction techniques that help forecast future trends or unknown values based on existing data.

1. Classification:

- Predicts categorical labels (e.g., spam vs. not spam).
- Techniques: Decision Trees, Naïve Bayes, SVM, KNN.

2. Regression:

- Predicts continuous values (e.g., sales, temperature).
- Types: Linear, Logistic, Polynomial Regression.

3. Time Series Forecasting:

- Predicts future values based on past time-dependent data.
- Uses: ARIMA, exponential smoothing.

4. Neural Networks:

- Deep learning models to predict complex patterns.
- Used for image classification, speech recognition.

5. Ensemble Methods:

- Combines multiple models for better prediction (e.g., Random Forest, Boosting).

These techniques are chosen based on data type, domain, and the prediction goal.

38. What do you mean by Non-linear Regression? In order to predict the probability of a student for getting into placement features are grade-point, communication skill and aptitude skill. What type of regression will you follow and why?

Answer:

Non-linear Regression:

- Used when the relationship between the dependent and independent variables is **non-linear**.
 - The model cannot be represented as a straight line.
-

Placement Prediction:

- The problem is to **predict probability** of placement (a classification or probability output).
- Appropriate technique: **Logistic Regression**.

Why Logistic Regression?

- It is used when the output is a **probability between 0 and 1**.
 - Fits a sigmoid curve (non-linear S-shape).
 - Models binary outcomes based on predictors like CGPA, communication, and aptitude.
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39. Explain the terms (a) True Positive (b) True Negative (c) False Positive (d) False Negative giving a suitable example.

Answer:

Let's consider a **medical test for a disease** as an example:

True Positive (TP)

o A case where the model correctly predicts a positive outcome.

o Example: In a medical test for detecting COVID-19, a true positive means the test correctly identifies an infected patient as positive.

2. True Negative (TN)

o A case where the model correctly predicts a negative outcome.

o Example: In the COVID-19 test scenario, a true negative means the test correctly identifies a healthy person as negative.

3. False Positive (FP) (Type I Error)

o A case where the model incorrectly predicts a positive outcome for a negative case.

o Example: The COVID-19 test incorrectly identifies a healthy person as positive, leading to unnecessary quarantine.

4. False Negative (FN) (Type II Error)

o A case where the model incorrectly predicts a negative outcome for a positive case.

o Example: The COVID-19 test fails to detect an infected patient, allowing them to spread the virus unknowingly.

Confusion Matrix:

	Predicted: Yes	Predicted: No
Actual: Yes	TP	FN
Actual: No	FP	TN

40. State and explain Bayes' theorem. Provide an example of its application.

Answer:

Bayes' Theorem

Bayes' Theorem is used to determine the conditional probability of an event. **It is used to find the probability of an event, based on prior knowledge of conditions that might be related to that event.**

Bayes' Theorem and Conditional Probability

Bayes theorem (also known as the Bayes Rule or Bayes Law) is used to determine the conditional probability of event A when event B has already occurred.

The general statement of Bayes' theorem is "**The conditional probability of an event A, given the occurrence of another event B, is equal to the product of the event of B, given A and the probability of A divided by the probability of event B.**" i.e.

$$P(A|B) = P(B|A)P(A) / P(B)$$

where,

- **P(A)** and **P(B)** are the probabilities of events A and B
- **P(A|B)** is the probability of event A when event B happens
- **P(B|A)** is the probability of event B when A happens

example

For example, if we want to find the probability that a white marble drawn at random came from the first bag, given that a white marble has already been drawn, and **there are three bags each containing some white and black marbles**, then we can use Bayes' Theorem.

Example 1: A person has undertaken a job. The Probabilities of completion of the job on time with and without rain are 0.44 and 0.95 respectively. If the probability that it will rain is 0.45, then determine the probability that the job will be completed on time.

- Let E_1 be the event that the mining job will be completed on time and E_2 be the event that it rains. We have,
- $P(A) = 0.45$,
 $P(\text{no rain}) = P(B) = 1 - P(A) = 1 - 0.45 = 0.55$
- By multiplication law of probability,
 $P(E_1) = 0.44$, and $P(E_2) = 0.95$
- Since, events A and B form partitions of the sample space S, by total probability theorem, we have
- $P(E) = P(A) P(E_1) + P(B) P(E_2)$
 $\Rightarrow P(E) = 0.45 \times 0.44 + 0.55 \times 0.95$
 $\Rightarrow P(E) = 0.198 + 0.5225 = 0.7205$
- So, the probability that the job will be completed on time is 0.7205

44. Why clustering is done? Compare and contrast various clustering algorithms.

Answer:

Purpose of Clustering:

- Clustering is an **unsupervised learning** technique used to group similar data points together.
- Useful in customer segmentation, anomaly detection, document classification.

Comparison of Algorithms:

Algorithm	Type	Strengths	Weaknesses
K-Means	Partitioning	Simple, scalable, fast	Needs K, sensitive to outliers
K-Medoids	Partitioning	Robust to outliers	More computationally expensive
DBSCAN	Density-based	Finds arbitrary shapes, handles noise	Poor for varying densities
Hierarchical	Hierarchical	No need for K, dendrogram output	Not scalable for large datasets
Mean Shift	Centroid	Doesn't require K, finds cluster shape	High computational cost

45. Why it is difficult to cluster data in high dimensional spaces? Which clustering method is best for high dimensional data?

Answer:

Challenges in High-Dimensional Data:

1. **Curse of Dimensionality:**
 - Distance measures become less meaningful as dimensions increase.
 - Data becomes sparse and scattered.
 2. **Computational Complexity:**
 - Memory and processing power requirements grow exponentially.
 3. **Noise and Irrelevant Features:**
 - High-dimensional data may contain many irrelevant or noisy features.
-

Best Clustering Method:

- **Subspace Clustering:** Discovers clusters in subspaces of the feature space.
 - **Spectral Clustering:** Handles high-dimensional similarity graphs well.
 - **BIRCH:** Efficient for large datasets with high dimensions.
 - **Autoencoders + K-Means:** First reduce dimensions, then cluster.
-

46. Which distance function is best for high dimensional data? How to handle high dimensional data?

Answer:

Best Distance Function:

- **Cosine Similarity:**
 - Focuses on the angle, not magnitude.
 - Effective when magnitude isn't important (e.g., document vectors).
- **Jaccard Distance:**
 - Useful for binary or sparse data.

Why not Euclidean?

- In high dimensions, Euclidean distance becomes less informative.
-

Handling High-Dimensional Data:

1. **Dimensionality Reduction:**
 - Techniques: PCA, t-SNE, Autoencoders.

2. **Feature Selection:**
 - Remove irrelevant or redundant features using information gain or correlation.
 3. **Regularization:**
 - Penalize large feature weights (L1, L2 regularization).
 4. **Clustering on Projections:**
 - Cluster on meaningful projections or subspaces.
-

48. Describe the pros and cons of K-means in comparison with K-medoid algorithm.

Answer:

Criteria	K-Means	K-Medoids
Centroid Type	Mean of points	Actual data point (medoid)
Robustness	Sensitive to outliers	Robust to noise and outliers
Computation	Faster, simple to implement	Slower, especially for large datasets
Use Case	Best for spherical clusters	Good for non-spherical or noisy data
Distance	Minimizes squared Euclidean distance	Minimizes dissimilarity measure

Conclusion:

- K-Means is preferred for speed and clean data.
 - K-Medoids is better for real-world data with noise or categorical features.
-

52. Illustrate the Hierarchical Clustering method.

Answer:

Hierarchical Clustering builds a multilevel hierarchy of clusters by either:

- **Agglomerative (Bottom-Up)** – start with individual points, merge step-by-step.
 - **Divisive (Top-Down)** – start with all data, divide recursively.
-

Steps in Agglomerative Clustering:

1. Start with **n** clusters (each point is its own cluster).
 2. Compute **distance matrix**.
 3. Merge the two closest clusters.
 4. Update the matrix.
 5. Repeat until all points are in one cluster.
-

Linkage Criteria:

- **Single Linkage:** Nearest neighbor
 - **Complete Linkage:** Farthest neighbor
 - **Average Linkage:** Average distance
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Dendrogram:

- A tree diagram showing merge/split operations.
- Helps choose number of clusters by cutting at a specific height.

Applications:

- Gene expression data, document clustering, customer segmentation.